Overview of evapotranspiration

How to compute evapotranspiration?

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2026-10-07

What is evapotranspiration?

What is evapotranspiration?

Definition

Defined as: *The combined processes through which water is transferred to the atmosphere from open water and ice surfaces, bare soil and vegetation that make up the Earth's surface.* ("Glossary" 2023)

Unit

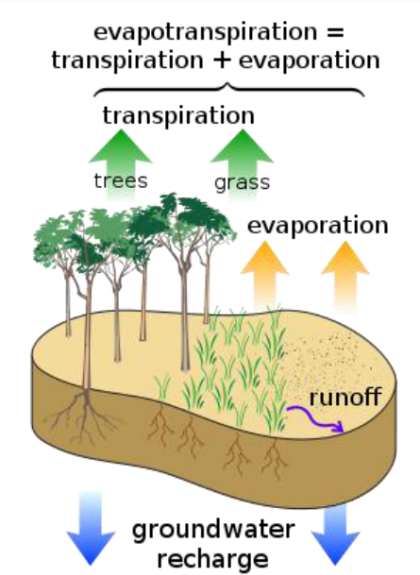
Typically measured in millimeters of water in a set unit of time, i.e. volume of water moved per unit area per unit of time.

- Millimeters per day: mm.day^{-1}
- Millimeters per month: mm.month^{-1}
- Millimeters per year: mm.year^{-1}

Key components of evapotranspiration

- **Evaporation (E)**: The physical process by which a liquid (e.g., water) becomes a gas (e.g., water vapour) below its boiling point.(“Glossary” 2023)
- **Transpiration (T)**: The process by which water vapor is released by plants through small openings called *stomata*.

$$ET = E + T$$



Focus on evaporation

Definition

Direct water movement into the air from

- Soil
- Water bodies
- Wet vegetation surfaces
- Intercepted rainfall

Driven factors

- temperature,
- humidity,
- solar radiation,
- wind.

Focus on transpiration

Water movement traveling from root systems through plants and exiting as vapor through tiny, microscopic pores on the outer surface of the leaves, called **stomata**.

Three main steps

1. Roots uptake water from the soil
2. Water moves through plant
3. Leaves release water vapor into the air through their stomata

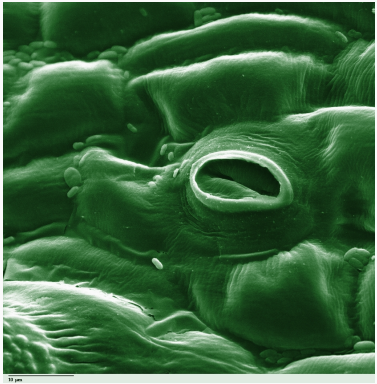


Figure 2: Stomata (Photohound, Public domain, via Wikimedia Commons)

Why do plants transpire?

- Cooling effect
- Nutrient transport
- Maintenance of internal cell pressure
- Opening of stomata, necessary for gas exchange during photosynthesis

How much water do plants transpire to produce dry matter?

$$\text{Transpiration ratio} = \frac{\text{Mass of water transpired}}{\text{Mass of dry matter produced}}$$

Typically ranging from 200 to 1000 kg of water for every kg of dry matter in crops

What factor affecting transpiration?

- Type of plant
- Soil type
- Solar radiation
- Precipitation and soil moisture
- Humidity
- Temperature
- Wind
- Land slope

What is the role of evapotranspiration?

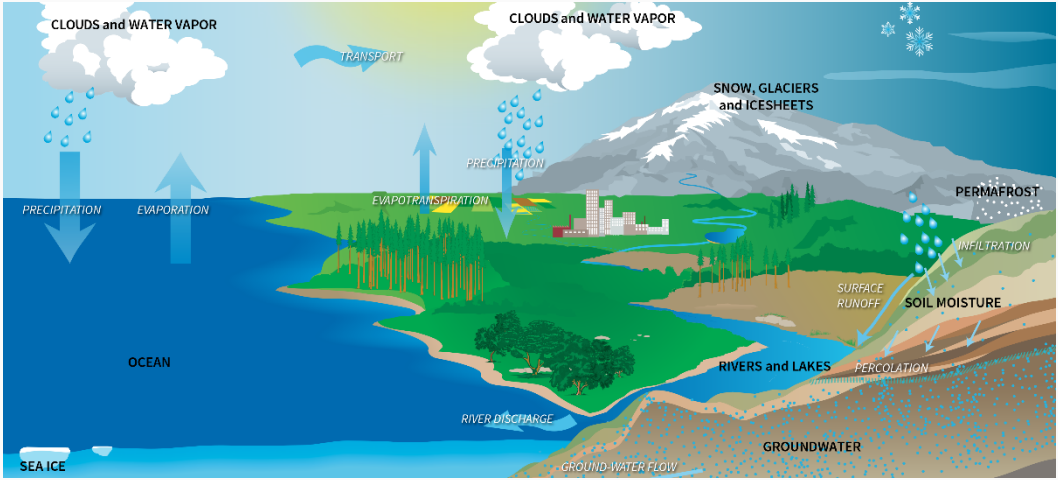
Major component of the hydrological cycle and climate

- Transfer water from the land surface to the atmosphere $\longrightarrow$ 2/3 of the precipitation
- Influence local and regional climate

Key role in water resource management and agricultural irrigation

- Determine crop water requirements
- Support drought monitoring and water-resource management

The water cycle and evapotranspiration



What are the primary drivers of evapotranspiration?

Water Availability The volume of water present in the soil and larger surface water bodies $\longrightarrow$ *fuel*

Available energy The amount of energy available in the air and soil $\longrightarrow$ *engine*

Atmospheric Demand The humidity levels and the atmosphere's capacity to absorb water $\longrightarrow$ *pump*

What are the impact of each meterological factors?

- **Solar radiation** Increase available energy
- **Air temperature** Higher temperatures generally increase atmospheric demand for water vapor.
- **Relative humidity** and **vapor-pressure deficit** High VPD usually increases ET if water is available.
- **Wind speed** Wind removes moist air near the surface and replaces it with drier air, increasing ET.
- **Precipitation** Rainfall supplies water for evaporation and replenishes soil moisture.

What are the impact of each surface/vegetation factors?

- **Soil moisture** Controls the variability and proportion of evaporation versus transpiration. When soil moisture becomes limited, actual ET declines.
- **Leaf area index** Higher LAI generally increases transpiration until the canopy becomes sufficiently dense.
- **Canopy resistance** High canopy resistance reduces transpiration.
- **Surface roughness** Tall or dense vegetation increases turbulence and can enhance exchange with the atmosphere.
- **Albedo** Bright surfaces reflect more solar radiation and generally absorb less energy.

What are the environmental and human factors?

- Topography and elevation
- Slope and aspect
- Soil texture
- Land-cover type
- Irrigation
- Drainage
- Salinity
- Urbanization
- Deforestation
- Fire and land degradation
- Seasonal vegetation development

Why are there different types of evapotranspiration?

- Reference evapotranspiration: ET_0
- Potential evapotranspiration: ET_p
- Crop evapotranspiration: ET_c
- Actual evapotranspiration: ET_a

Reference evapotranspiration: ET_0

- Represents the atmospheric evaporative demand for a standardized, well-watered reference surface.
- Uses a hypothetical grass reference surface
  - A short actively growing grass, about 0.12 m tall, completely shading the ground, well supplied with water, with standardized surface resistance and albedo
- Depends mainly on weather conditions: air temperature, solar radiation, wind speed, humidity, atmospheric pressure
- Provides a ET baseline based on atmospheric demands (meteorological conditions)

Potential evapotranspiration: ET_p

- Represents evapotranspiration that could occur if water were freely available.
- Assumes that:
  - Soil water content is not limiting
  - The surface can evaporate and transpire at its maximum rate
  - Atmospheric conditions determine the evaporative demand
- Refers to maximum possible ET

Crop evapotranspiration: ET_c

- Represents evapotranspiration of a particular crop under adequate water supply and non-stressed conditions.

$$ET_c = K_c ET_0$$

where:

- ET_c = crop evapotranspiration
- K_c = crop coefficient $\rightarrow$ **Crop characteristics**
- ET_0 = reference evapotranspiration $\rightarrow$ **Atmospheric demand**

Crop evapotranspiration: crop coefficient

- Depends on: crop type, crop height, growth stage, canopy development, surface roughness
- Varies during crop development

1. **Initial stage** Small plants and substantial exposed soil
   $\rightarrow$ low K_c
2. **Development stage** Increasing canopy and transpiration
   $\rightarrow$ increasing K_c
3. **Mid-season stage** Full canopy and maximum water use
   $\rightarrow$ high K_c
4. **Late-season stage** Senescence and reduced transpiration
   $\rightarrow$ decreasing K_c

Actual evapotranspiration: ET_a

- Represents the amount of water that actually leaves a real soil–plant system.
- Depends on:
  - Weather conditions
  - Soil-water availability
  - Crop type and growth stage
  - Canopy characteristics
  - Irrigation and management
  - Plant water stress

$$ET_a < ET_p$$

Relationships between ET terms

$$ET_0 \longrightarrow ET_c \longrightarrow ET_a$$

How to take into account for water stress?

If water is limited, actual ET is lower than unstressed crop ET.

Single-coefficient approach

$$ET_a = K_s ET_c = K_s K_c ET_0$$

where:

- K_s = water-stress coefficient $0 \leq K_s \leq 1$
  - $K_s = 1$: no water stress
  - $K_s < 1$: water stress reduces ET
  - $K_s = 0$: ET is effectively stopped in the model

How to take into account for water stress?

If water is limited, actual ET is lower than unstressed crop ET.

Dual-coefficient approach

$$ET_a = (K_{cb}K_s + K_e)ET_0$$

where:

- K_{cb} = basal crop transpiration coefficient
- K_s = water-stress coefficient
- K_e = soil-evaporation coefficient

The dual approach separates:

- Transpiration by the crop
- Evaporation from the soil surface

Summary table

Concept	Main question answered
ET_0	How strong is the atmospheric evaporative demand?
ET_p	How much ET could occur if water were unlimited?
ET_c	How much would this crop use water if it were not stressed?
ET_a	How much water is actually being lost?

Key equations

$$ET = E + T$$

$$ET_c = K_c ET_0$$

$$ET_a = K_s K_c ET_0$$

or, with separate soil evaporation:

$$ET_a = (K_{cb} K_s + K_e) ET_0$$

How to model evapotranspiration?

Two approaches

- Water Balance Equation
- Surface Energy Balance Equation

Water Balance Equation

$$P = ET + R + D + \Delta S$$

Where:

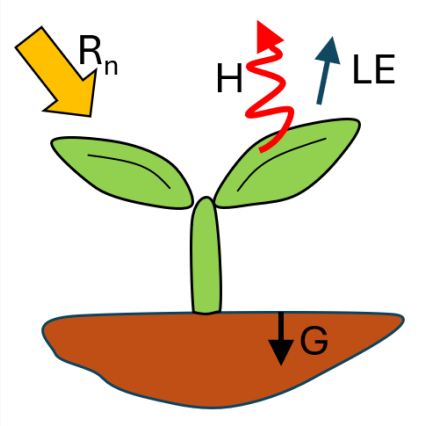
- P = precipitation
- ET = evapotranspiration
- R = runoff
- D = deep drainage or groundwater recharge
- ΔS = change in water storage

Surface energy equation equation

$$R_n = G + LE + H$$

Where

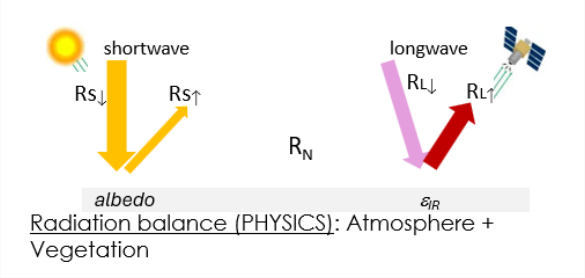
- R_n = net radiation
- G = soil heat flux
- H = sensible heat flux
- LE = latent heat flux



$$LE = L \times ET$$

where L is the specific latent heat for vaporization

Net radiation



Radiation balance (PHYSICS): Atmosphere + Vegetation

$$R_n = R_S^{\downarrow} - R_S^{\uparrow} + R_L^{\downarrow} - R_L^{\downarrow}$$

$$R_n = R_S^{\downarrow}(1-\alpha) + \epsilon(R_L^{\downarrow} - \sigma_{SB} T_s^4)$$

Where

- $R_S^{\downarrow}$ shortwave downwelling radiation
- $R_L^{\downarrow}$ longwave downwelling radiation
- ϵ land surface emissivity
- α land surface albedo
- T_s land surface temperature
- σ_{SB} Stefan-Boltzmann constant

How to represent the soil-vegetation system?

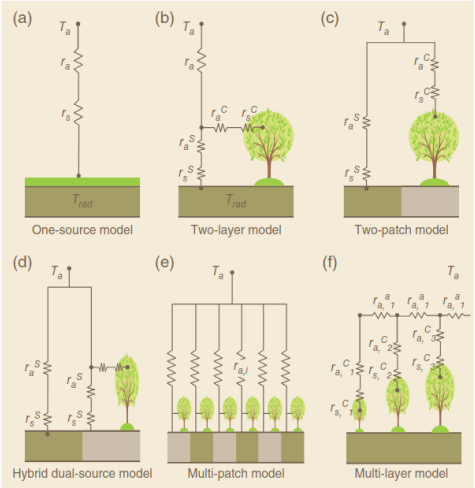


Figure 3: Different configurations of resistance networks(Zhang et al. 2016)

How to represent the soil-vegetation system?

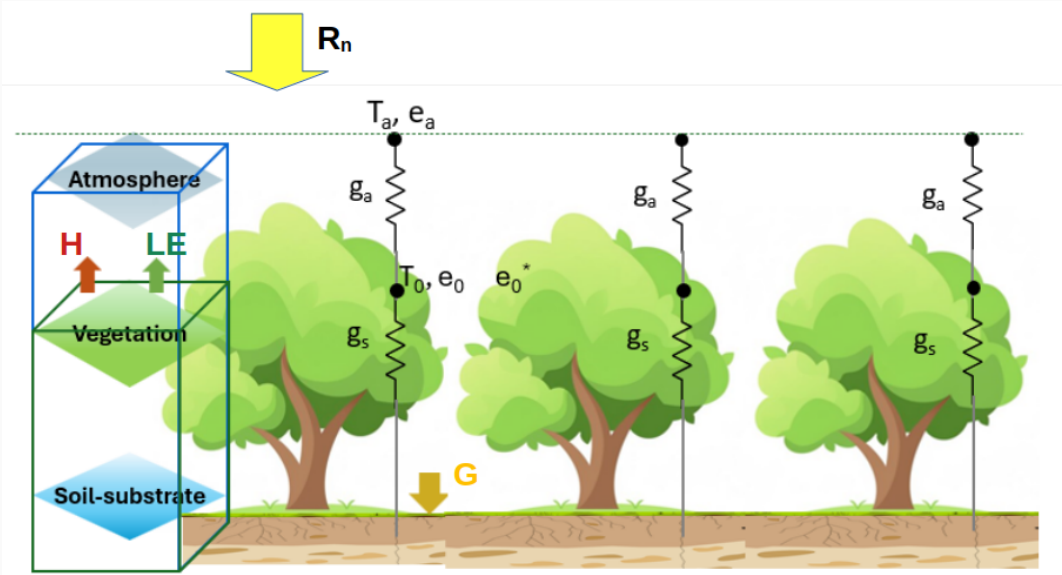
Air resistance / air conductance

$$r_a = \frac{1}{g_a}$$

Stomatal resistance / stomatal conductance

$$r_s = \frac{1}{g_s}$$

Focus with one source model



What temperature are we talking about?

- **Aerodynamic temperature**: Aerodynamic temperature is the air temperature at the thermal roughness length. It is a key parameter in models simulating land-atmosphere processes, which are relevant in numerous applications.
- **Thermodynamic temperature**: Thermodynamic temperature is a macroscopic quantity thought to be constant throughout any group of subsystems that are in thermodynamic equilibrium, assuming therefore no heat transfer.
- **Directional brightness temperature**: It is the temperature of a black body that would have the same radiance as the radiance actually observed with the radiometer.
- **Directional radiometric temperature**: A directional radiometric temperature is derived from a radiative energy balance of a surface and provides the best approximation to the thermodynamic temperature based on a measure of radiance. It is the quantity to estimate from satellite observations in the domain of LST inversions.

Analogy with Ohm's law

$$\text{Flux} = \frac{\text{Difference in potential}}{\text{Resistance}} = \text{Conductance} \times (\text{Difference in potential})$$

- **Electricity** $I = U/R = \Delta V/R$ (Ohm's law) The flux of electric current is directly related to the difference in electric potential.
- **Thermodynamics** $\Phi = -k\nabla T$ (Fourier's law) Heat flux flows from regions of high thermal potential (high temperature) to regions of low potential (low temperature)
- **Fluid mechanics** Fluid flux flows from areas of high pressure (high pressure potential) to areas of low pressure.

Air conductance

- Represents **how easily water vapor can move through the air layer surrounding the leaf**
- Depends on:
  - **Wind speed** $\longrightarrow$ stronger wind removes the moist air layer around the leaf, allowing more evaporation
  - **Roughness** $\longrightarrow$ Leaf shape and size change the roughness and affect how turbulent the air becomes
- Is largely passive and not controlled by the plant

Stomatal resistance

- Represents **how open the stomata are to allow water vapor to escape**
  - When stomata are wide open, conductance is high $\Rightarrow$ water vapor escapes easily from inside the leaf
  - When stomata are closed or partially closed, conductance is low $\Rightarrow$ less water vapor can escape
- Plants control the stomata opening:
  - they open stomata to take in CO_2 for photosynthesis, but this comes at the cost of water loss.
  - they close stomata to conserve water, especially during dry conditions.

Air conductance and stomatal conductance in evapotranspiration

- If stomatal conductance is very low but air conductance is high, water loss is still limited by the stomata.
- If stomatal conductance is high but air conductance is low (still air), water loss is limited by how slowly vapor can diffuse away.

For one source representation

$$H = \rho c_p g_a (T_0 - T_a)$$

where

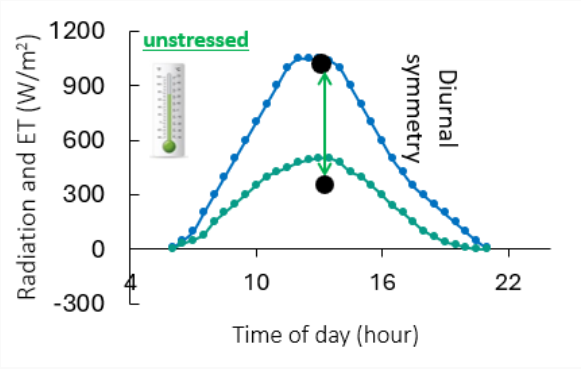
- ρ is the density of air at constant pressure ($kg.m^{-3}$),
- c_p is the specific heat of air at constant pressure ($J.kg^{-1}.K^{-1}$) $\rightarrow$ *It corresponds to the amount of heat that must be added to one unit of mass of air in order to cause an increase of one unit in temperature.*
- T_0 is the aerodynamic temperature
- T_a is the air temperature
- g_a air conductance ($m.s^{-1}$)

$$LE = \frac{\rho c_p}{\gamma} \frac{g_a g_s}{g_a + g_s} (e_0^\star - e_a)$$

where

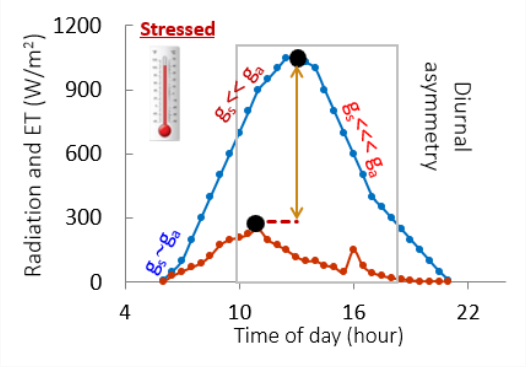
- ρ is the density of air at constant pressure ($kg.m^{-3}$),
- c_p is the specific heat of air at constant pressure ($J.kg^{-1}.K^{-1}$)
- γ is the psychrometric constant ($hPa.K^{-1}$) $\rightarrow$ It relates the partial pressure of water in air to the air temperature.
- $e_0^\star$ is the saturation vapor pressure at T0 (hPa)
- e_a is the vapor pressure at Ta (hPa)
- g_a air conductance ($m.s^{-1}$)
- g_s stomatal conductance ($m.s^{-1}$)

Ecosystem functioning with unstressed stomata



- Radiation explains **75 - 80%** of ET variability
- Stomatal control: **not significant**

Stressed vegetation



- **strong hysteresis**: Surface temperature vs radiation
- **high thermal inertia**

A very large numbers of evapotranspiration methods

Overview of evapotranspiration estimation approaches

1. Empirical or meteorological models
2. Physically based surface-energy-balance models
3. Models using optical remote-sensing data
4. Models using optical and thermal infrared remote-sensing data

Each method involves a trade-off between

- Data requirements
- Physical realism
- Spatial coverage
- Temporal frequency
- Computational complexity
- Accuracy and uncertainty

Empirical and Meteorological Models

Typical inputs

- Temperature, Precipitation, Solar radiation, Relative humidity, Wind speed, Day length

Advantages

- Easy to implement
- Low computational cost
- Useful where data are limited
- Suitable for regional or long-term studies

Limitations

- Often calibrated for specific climates
- May not transfer well to other regions
- Can perform poorly under unusual conditions
- May not represent vegetation stress explicitly

Physically based surface-energy-balance models: Penman-Monteith

$$LE = \frac{\Delta(R_n - G) + \rho C_p g_a VPD}{\Delta + \gamma(1 + g_a/g_s)}$$

with $\Delta = \frac{\mathrm{d}e_{sat}}{\mathrm{d}T}$ and $VPD = e_{sat} - e_a$

- e_{sat} saturated vapor pressure
- e_a vapor pressure at air temperature
- VPD Vapor Pressure Deficit of air
- γ psychrometric constant
- g_a aerodynamic conductance modeled based on crop height, u wind, roughness parameters
- g_s surface conductance modeled based on LAI, leaf resistance

Use to compute reference

$$ET_0 = \frac{0.408\Delta(R_n - G) + \gamma\frac{900}{T_a}uVPD}{\Delta + \gamma(1 + 0.34u)}$$

- T_a air temperature
- u wind
- Δ slope of the saturation vapor pressure and temperature curve
- VPD vapor pressure deficit of air
- R_n net radiation
- G ground heat flux

See FAO56 (Allen et al. 1998)

Physically based surface-energy-balance models: FAO56

Advantages

- Physically based
- Widely accepted for reference ET
- Performs well when complete weather data are available

Limitations

- Requires many meteorological inputs
- Sensitive to the quality of radiation, wind, humidity, and temperature data
- Reference-surface assumptions may not describe heterogeneous landscapes

Physically based surface-energy-balance models: Priestley-Taylor

$$LE = \frac{\alpha_{PT}\Delta(R_n - G)}{\Delta + \gamma}$$

- α_{PT} Priestley-Taylor parameter
- Do not take into account the water stress

Models using optical remote-sensing data

Some models combine optical data with radiation and meteorological observations.

Optical remote sensing provides information about:

- Albedo
- Fraction of vegetation cover
- LAI
- Canopy structure
- Surface roughness
- Photosynthetic activity

Optical remote sensing does not directly measure ET. Instead, it estimates variables related to transpiration and vegetation condition.

Models using optical remote-sensing data

Advantages

- Relatively simple
- Useful for agricultural monitoring
- Provides spatial information
- Can be applied over large areas

Limitations

- NDVI can saturate in dense vegetation
- Optical observations are affected by clouds
- Vegetation indices do not directly measure water stress
- Calibration is often crop- and region-dependent

Interest of surface temperature

Land Surface Temperature is useful because ET cools the land surface:

- High ET usually produces lower surface temperature
- Low ET or water stress often produces higher surface temperature

TIR remote sensing is therefore particularly useful for detecting water stress.

Surface temperature is a key indicator of *plant water stress* and *soil moisture conditions.*

What Does a Hot Surface Indicate?

A *hot surface* may indicate:

- Low soil moisture
- Reduced transpiration
- Closed stomata
- Sparse vegetation
- High sensible heat flux
- Strong atmospheric demand

What Does a Cool Surface Indicate?

A *cool surface* may indicate:

- Active transpiration
- Wet soil
- Dense vegetation
- Irrigation
- High latent heat flux

Factors Influencing Surface Temperature

However, **surface temperature** can also be influenced by:

- Wind
- Surface roughness
- Air temperature
- Radiation
- Soil texture
- Vegetation structure

Interpreting Surface Temperature

Therefore, temperature should be interpreted together with meteorological and vegetation data.

$$LE = R_n - G - H$$

$$H = \rho C_p \frac{T_O - T_a}{r_a}$$

- ρ air density
- C_p specific heat
- T_0 aerodynamic surface temperature $T_0 \approx T_s$
- r_a aerodynamic resistance modeled based on crop height, u wind, roughness parameters

Single-Source Energy-Balance Models: General procedure

1. Estimate surface temperature
2. Calculate net radiation
3. Estimate soil heat flux
4. Estimate sensible heat flux
5. Derive latent heat flux
6. Convert latent heat flux to ET

Single-Source Energy-Balance Models

Strengths

- Provides spatially distributed ET
- Detects field-scale water stress
- Useful for irrigation management
- Can cover large regions

Limitations

- Requires careful calibration
- Sensitive to atmospheric data

Single-Source Energy-Balance Models: Examples

- SEBAL Surface Energy Balance Algorithm for Land
- METRIC Mapping Evapotranspiration at High Resolution with Internalized Calibration
- SEBS Surface Energy Balance System

Two-source models separate the land surface into: vegetation (or canopy) and soil

$$LE = R_n - G - H = R_n - G - H_{canopy} - H_{soil}$$

$$H_{canopy} = \rho C_p \frac{T_{canopy} - T_a}{r_{canopy}}$$

$$H_{soil} = \rho C_p \frac{T_{soil} - T_a}{r_{soil}}$$

$$T_s = \left(f_c T_{canopy}^4 + (1 - f_c) T_{soil}^4 \right)^{1/4}$$

- f_c vegetation fraction cover
- T_{canopy} canopy temperature and T_{soil} soil temperature
- r_{soil} surface resistance and r_{canopy} canopy resistance modeled based on LAI, leaf resistance

Two-Source Energy-Balance Models

Advantages

- Separates evaporation and transpiration
- Better represents sparse vegetation
- Useful for heterogeneous surfaces
- Can improve water-stress estimation

Limitations

- Requires more parameters
- More computationally complex
- Needs information about canopy and soil temperatures or their partitioning

Two-Source Energy-Balance Models: Examples

- ALEXI Atmosphere–Land Exchange Inverse
- DisALEXI Disaggregation Atmosphere–Land Exchange Inverse
- SPARSE Soil Plant Atmosphere and Remote Sensing Evapotranspiration

Comparison of ET Estimation Methods

Method	Main data	Main output	Strengths	Limitations
Empirical models	Various weather data	Potential or reference ET	Simple and inexpensive	Site-specific and less physically detailed
Penman–Monteith	Radiation, temperature, humidity, and wind	Reference ET	Physically based and widely accepted	Requires complete weather data
Optical remote sensing	Reflectance, vegetation indices, LAI, albedo	Spatial crop coefficients or vegetation parameters	Large-area vegetation monitoring	Weak direct sensitivity to water stress
TIR remote sensing	LST, radiation, weather data	Actual ET and surface fluxes	Detects water stress and spatial ET patterns	Cloud-sensitive and data-intensive
Hybrid models	Weather, optical, TIR, and land-surface data	High-resolution actual ET	Combines complementary information	More complex and requires integration

TRISHNA evapotranspiration models

Model selected for TRISHNA

- STIC **Surface Temperature Initiated Closure**
- EVASPA **Evapotranspiration from SPAce**

Why this choice ?

- Delivering actual evapotranspiration (ET) at less than 100m resolution is challenging, mainly because the performance is driven by meteorological inputs that are only available at a very low spatial resolution (0.1°) through ECMWF analysis.
- STIC require few meteorological data, only air temperature and dewpoint temperature and daily radiation
- EVASPA require only daily radiation

STIC: Surface Temperature Initiated Closure

Overview

Surface Temperature Initiated Closure (STIC) Algorithm (Mallick et al. 2014)

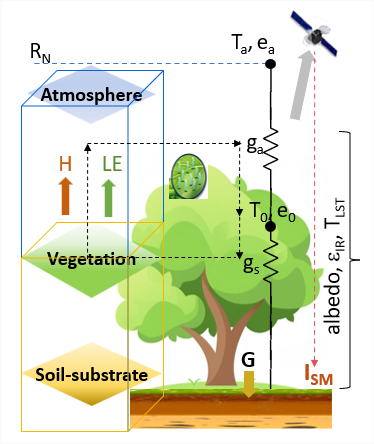
- Purpose: Single-pixel evapotranspiration (ET) estimation
- Model Type: One-source Surface Energy Balance (SEB) model
- Innovation: Uniquely integrates Land Surface Temperature (LST) directly into the Penman-Monteith formulation

Meteorological inputs

The objective of minimizing the impact of exogenous meteorological information on the performance of the product, as it only depends on

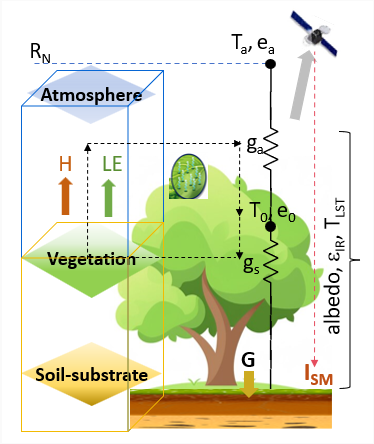
- **air temperature**
- **dewpoint temperature** (i.e. air humidity)

Analytical Framework



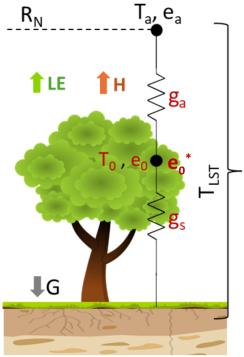
- Combines LST with aerodynamic and heat balance equations
- Determines two conductances:
  - Aerodynamic conductance (g_a) — efficiency of air transport of heat and moisture
  - Surface conductance (g_s) — vegetation canopy's ability to exchange water vapor

Analytical Framework



- Integrates:
  - LST-driven water stress index
  - Aerodynamic equations for sensible (H) and latent (LE) heat
  - Modified complementary relationship advection-aridity hypothesis
  - Shuttleworth-Wallace sparse canopy + Penman-Monteith big-leaf models

Principle



$$H = \rho c_p g_a (T_0 - T_a)$$

$$LE = \frac{\rho c_p}{\gamma} g_a (e_0 - e_a)$$

$$= \frac{\rho c_p}{\gamma} g_s (e_0^* - e_0)$$

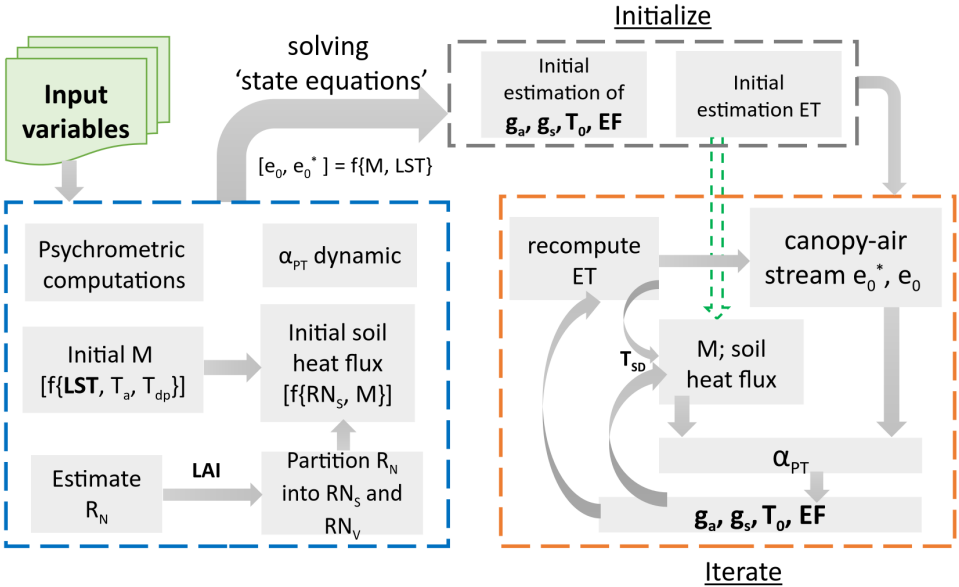
$$= \frac{\rho c_p}{\gamma} \frac{g_a g_s}{g_a + g_s} (e_0^* - e_a)$$

$$g_a = \frac{\gamma R_N - \rho c_p g_s (e_0^* - e_0)}{\gamma \rho c_p (T_0 - T_a)}$$

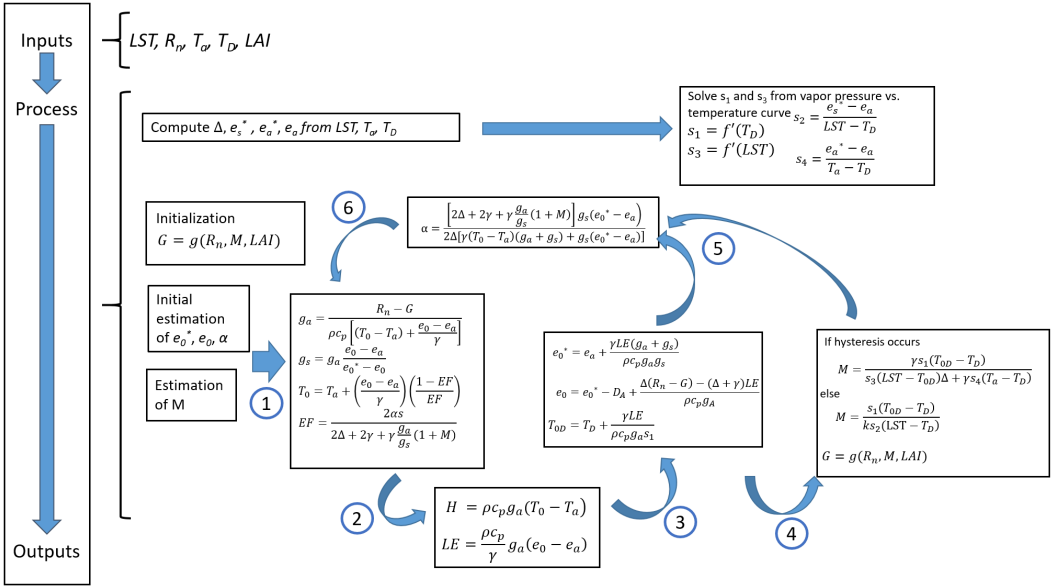
$$g_S = \frac{\gamma g_a \{R_N - H\}}{\rho c_p g_a (e_0^* - e_a) - \gamma \{R_N - H\}}$$

Dry ($T_0 >> T_a$)

Wet ($T_0 \approx T_a$)


Initialize
Input variables
solving 'state equations'
[e0, e0*] = f{M, LST}
Initial estimation of ga, gs, T0, EF
Initial estimation ET
Psychrometric computations
αPT dynamic
Initial M [f{LST, Ta, Tdp}]
Initial soil heat flux [f{RNs, M}]
Estimate RN
LAI
Partition RN into RNs and RNv
recompute ET
canopy-air stream e0*, e0
TSD
M; soil heat flux
αPT
ga, gs, T0, EF
Iterate

Detailed algorithm



EVASPA: EVApotranspiration from SPAce

Overview

Ensemble approach

- Uses multiple contextual algorithms based on EF models
- Computes average and dispersion of results across algorithms
- Provides final EF estimate and reliability assessment

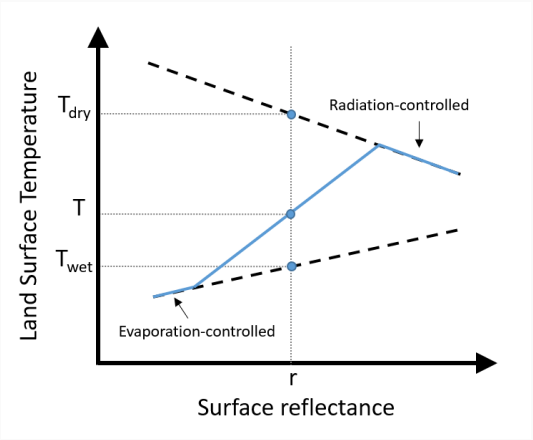
See (Gallego-Elvira et al. 2013)

Methodological principle: Model Assumptions & Limitations

- Requires only incident radiation data (no auxiliary meteorological inputs)
- Assumes land surface temperature variations result from variations in water content, vegetation, or surface albedo
- Assumes no variations in climatic variables, no variations in aerodynamic parameters, and no variations related to topography across the domain
- **Limitation:** Algorithm fails if these uniform conditions aren't met

Methodological principle

- Correlation between surface temperature and reflectance of areas subject to constant atmospheric forcing
- Two distinct thermal regimes exist under constant atmospheric forcing:
  - **Evaporation-controlled** (lower temperature): Temperature changes driven by soil moisture availability
  - **Radiation-controlled** (higher reflectance): Temperature decreases with increasing reflectance when soil moisture is depleted and evaporation ceases



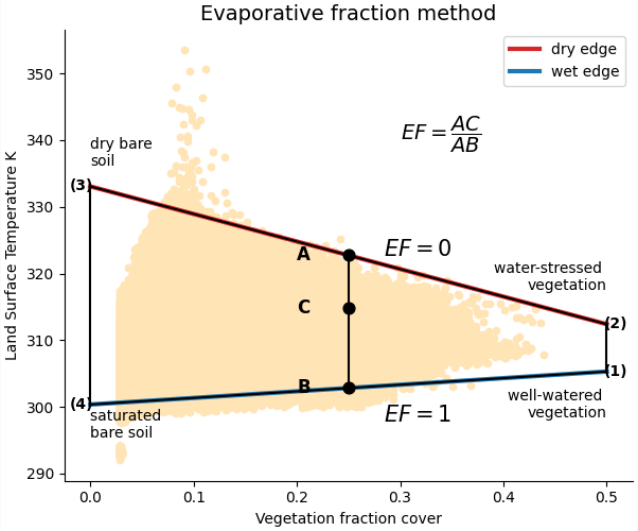
Algorithm: Evaporative fraction

Definition

- Quantity of energy which is allocated to evapotranspiration in regards to the available energy.
- Defined as

$$EF = \frac{LE}{LE + H} = \frac{LE}{R_n - G}$$

Algorithm: Evaporative fraction estimation



$$EF = \frac{T_{dry} - T}{T_{dry} - Twet}$$

- For fully wet conditions, i.e. T_{wet} where $LE = Rn - G$ and $H = 0$
- For fully dry conditions, i.e. T_{dry} where $H = Rn - G$ and $LE = 0$

Algorithm: Ensemble approach

1. Several methods are used to estimate dry and wet edges related to several LST vs variables
2. Each EF is estimated for each methods
3. EF is deduced by merging each EF

Algorithm: Evapotranspiration from evaporative fraction

$$ET = \frac{LE}{L} = \frac{EF(R_n - G)}{L}$$

How to extrapolate to daily

Extrapolation to daily evapotranspiration

- Done through an extrapolation of instantaneous data to the daily scale thanks to the use of the solar radiation ratio.
- Defined as

$$ET^{daily} = \frac{R_{SD}^{daily}}{R_{SD}^{inst}} ET^{inst}$$

- ET^{inst} evapotranspiration at the time of TRISHNA acquisition
- ET^{daily} ($mm.d^{-1}$) daily evapotranspiration
- R_{SD}^{inst} downwelling shortwave radiation at the time of TRISHNA acquisition
- R_{SD}^{daily} daily value of downwelling shortwave radiation

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